DATA ANALYTICS PROJECT

SUPERSTORE SALES ANALYSIS

Using Python and MS Excel

Date: 16/05/2026

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TOOLS USED FOR THIS DATASET

Python	MS Excel
• Data analysis	• Pivot tables
• Statistical operations	• Dashboard creation
• Visualization	• KPI analysis
	• Interactive reporting

PROJECT OVERVIEW

The dataset contains information about:

- Orders
- Customers
- Sales
- Profit
- Categories
- Regions
- Shipping modes

OBJECTIVES

- Identify top-performing customers
- Analyze sales and profit trends
- Find profitable regions and categories
- Analyze loss-making products
- Create interactive dashboards
- Generate business insights using analytics

QUESTIONS

- Which customers ordered most frequently?
- Enter a state name and display total sales and profit.
- Which orders generated the highest profit?
- Enter a customer name and display all orders.
- What is the standard deviation of sales and profit?
- What are the mean, median, maximum and minimum sales values?
- Which category generated the highest total sales?
- Which sub-category generated the highest profit?
- Which state has the highest sales?
- Which region is the most profitable?

CUSTOMER & PROFIT ANALYSIS

Top customers

```
top_customers = (df['Customer Name'].value_counts().head(10).reset_index())
top_customers.columns = ['Customer Name', 'Number of Orders']
top_customers
```

	Customer Name	Number of Orders
0	William Brown	37
1	Matt Abelman	34
2	John Lee	34
3	Paul Prost	34
4	Jonathan Doherty	32
5	Chloris Kastensmidt	32
6	Seth Vernon	32
7	Edward Hooks	32
8	Emily Phan	31
9	Zuschuss Carroll	31

Top profitable orders

```
top_profit_orders = (df[['Order ID', 'Customer Name', 'Sales', 'Profit']].sort_values(by='Profit', ascending=False).head(10))

top_profit_orders
```

	Order ID	Customer Name	Sales	Profit
6826	CA-2016-118689	Tamara Chand	17499.950	8399.9760
8153	CA-2017-140151	Raymond Buch	13999.960	6719.9808
4190	CA-2017-166709	Hunter Lopez	10499.970	5039.9856
9039	CA-2016-117121	Adrian Barton	9892.740	4946.3700
4098	CA-2014-116904	Sanjit Chand	9449.950	4630.4755
2623	CA-2017-127180	Tom Ashbrook	11199.968	3919.9888
509	CA-2015-145352	Christopher Martinez	6354.950	3177.4750
8488	CA-2016-158841	Sanjit Engle	8749.950	2799.9840
7666	US-2016-140158	Daniel Raglin	5399.910	2591.9568
6520	CA-2017-138289	Andy Reiter	5443.960	2504.2216

STATISTICAL ANALYSIS USING NUMPY

Standard Deviation

```
print("Sales Standard Deviation:",
np.std(df['Sales']))
print("Profit Standard Deviation:",
np.std(df['Profit']))
```

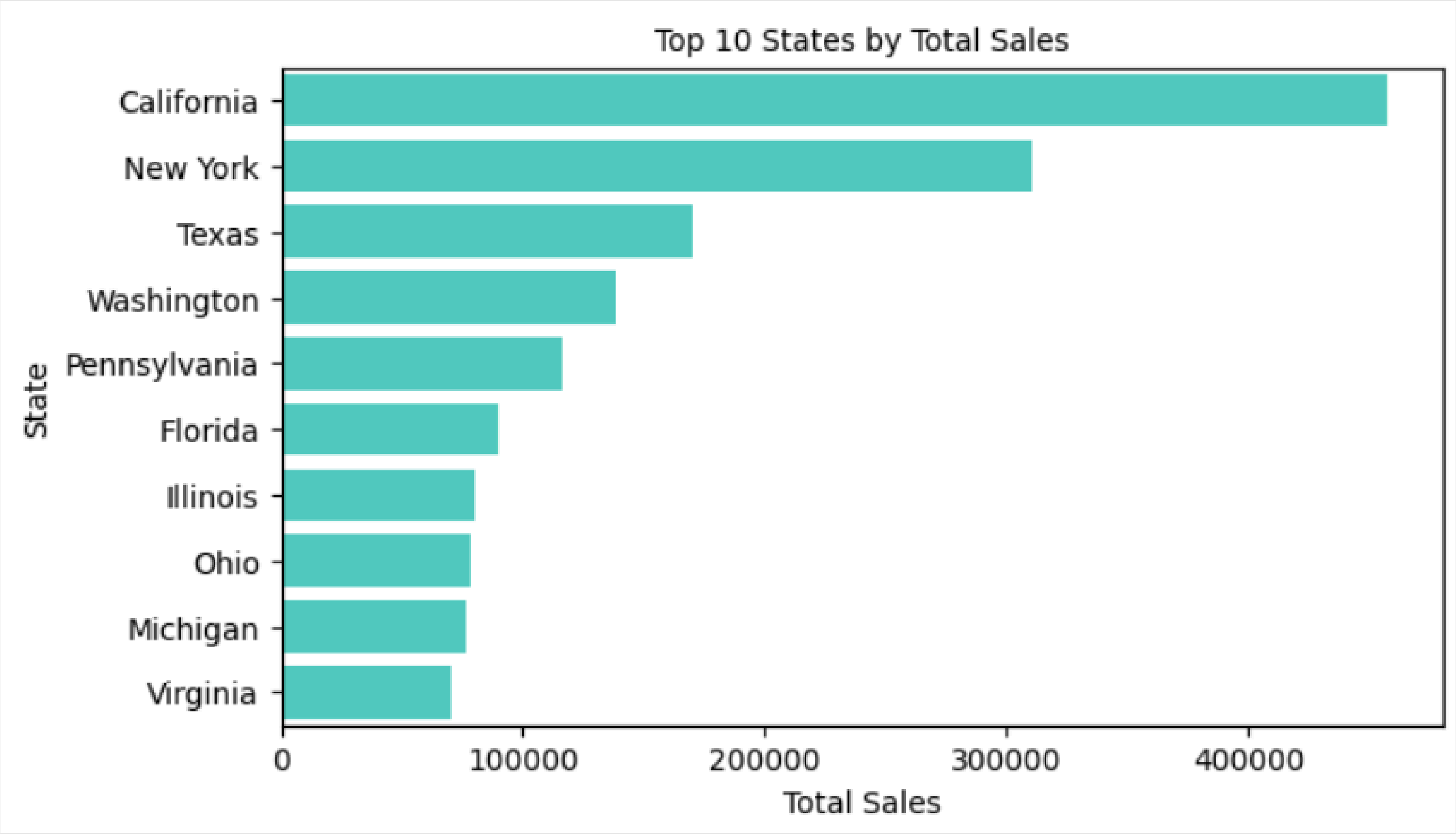
```
Sales Standard Deviation: 623.2139187650461
Profit Standard Deviation: 234.2483873603588
```

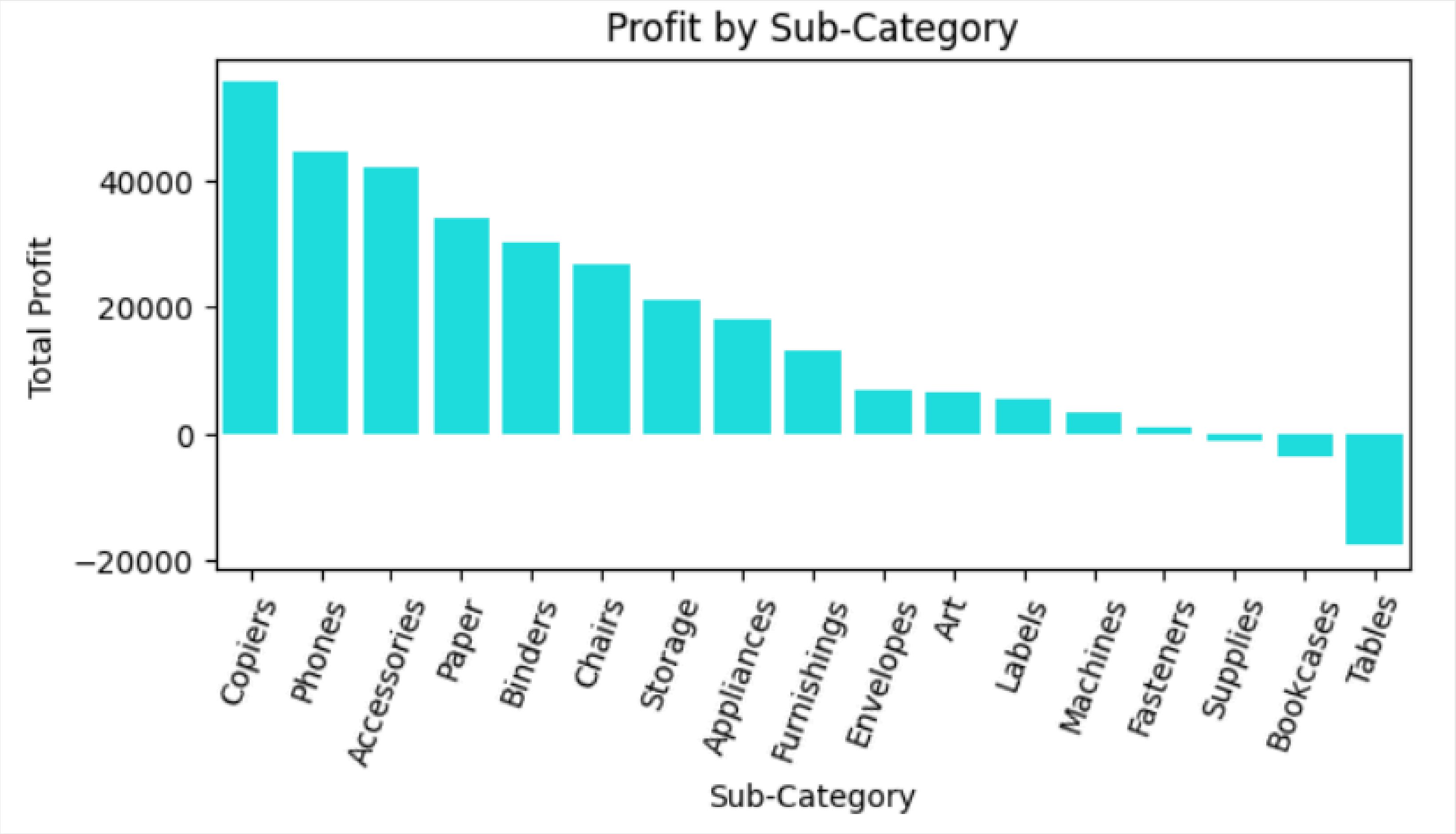
Mean, Median, Max, Min - Sales

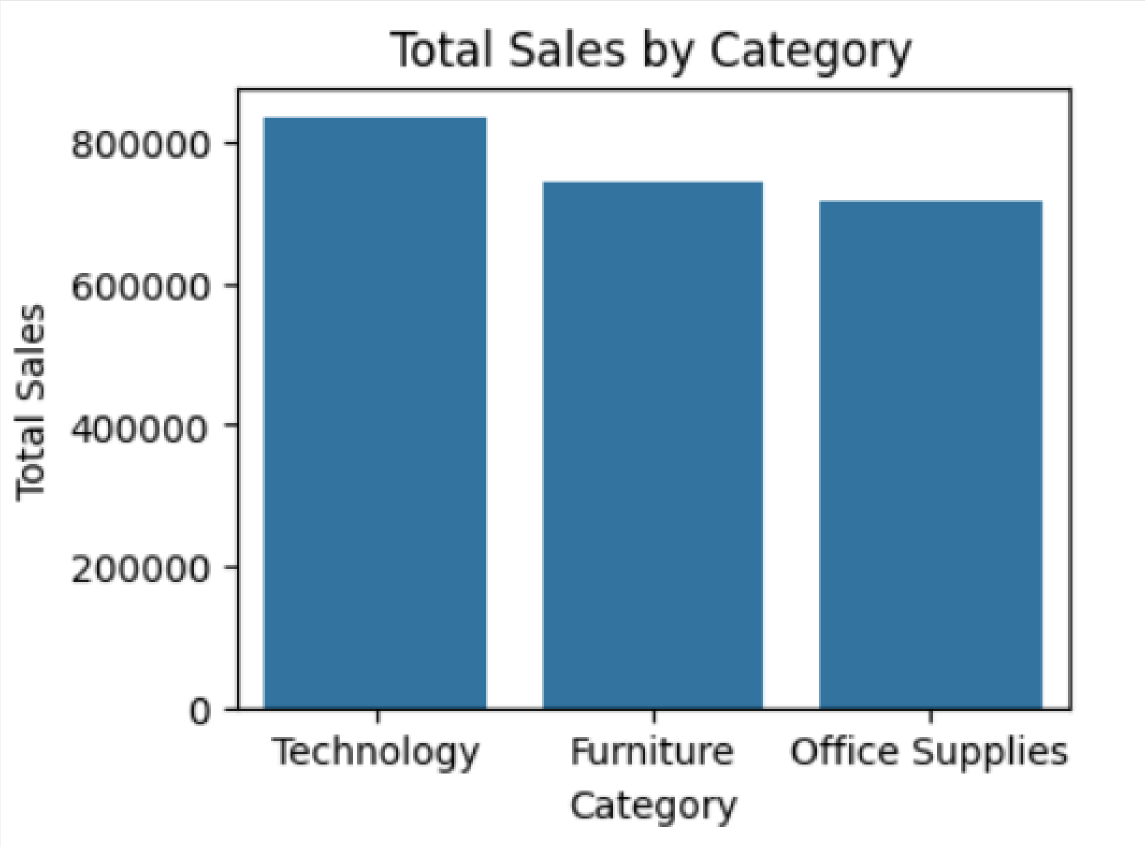
```
print("Mean Sales:", np.mean(df['Sales']))
print("Median Sales:", np.median(df['Sales']))
print("Maximum Sales:", np.max(df['Sales']))
print("Minimum Sales:", np.min(df['Sales']))
```

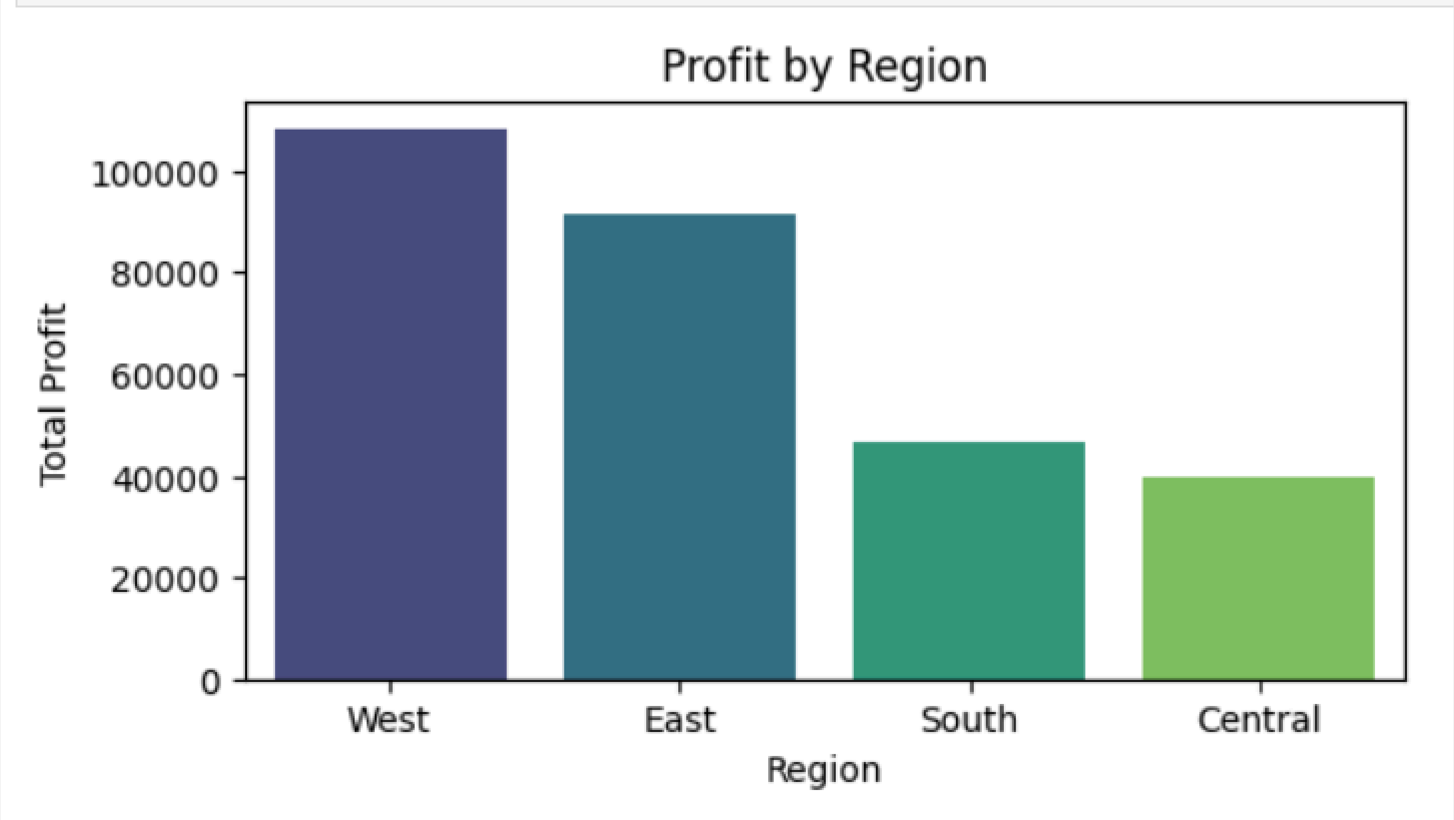
```
Mean Sales: 229.85800083049827
Median Sales: 54.489999999999995
Maximum Sales: 22638.48
Minimum Sales: 0.444
```

SALES AND PROFIT ANALYSIS









EXCEL ANALYSIS

PROFIT BY CUSTOMER SEGMENT

Segment	Total Profit
Consumer	$1,34,119
Corporate	$91,979
Home Office	$60,299
Grand Total	**$2,86,397**

MONTHLY REVENUE TREND

Month	Total Sales
Jan	$94,925
Feb	$59,751
Mar	$2,05,005
Apr	$1,37,762
May	$1,55,029
Jun	$1,52,719
Jul	$1,47,238
Aug	$1,59,044
Sep	$3,07,650
Oct	$2,00,323
Nov	$3,52,461
Dec	$3,25,294
Grand Total	**$22,97,201**

AVERAGE DISCOUNT BY CATEGORY

Row Labels	Average of Discount
Furniture	17.39%
Office Supplies	15.73%
Technology	13.23%
Grand Total	**15.62%**

LOSS-MAKING SUB-CATEGORIES

Sub-Cate	Total Loss
Tables	-$17,725.48
Bookcases	-$3,472.56
Supplies	-$1,189.10
Grand Total	**-$22,387.14**

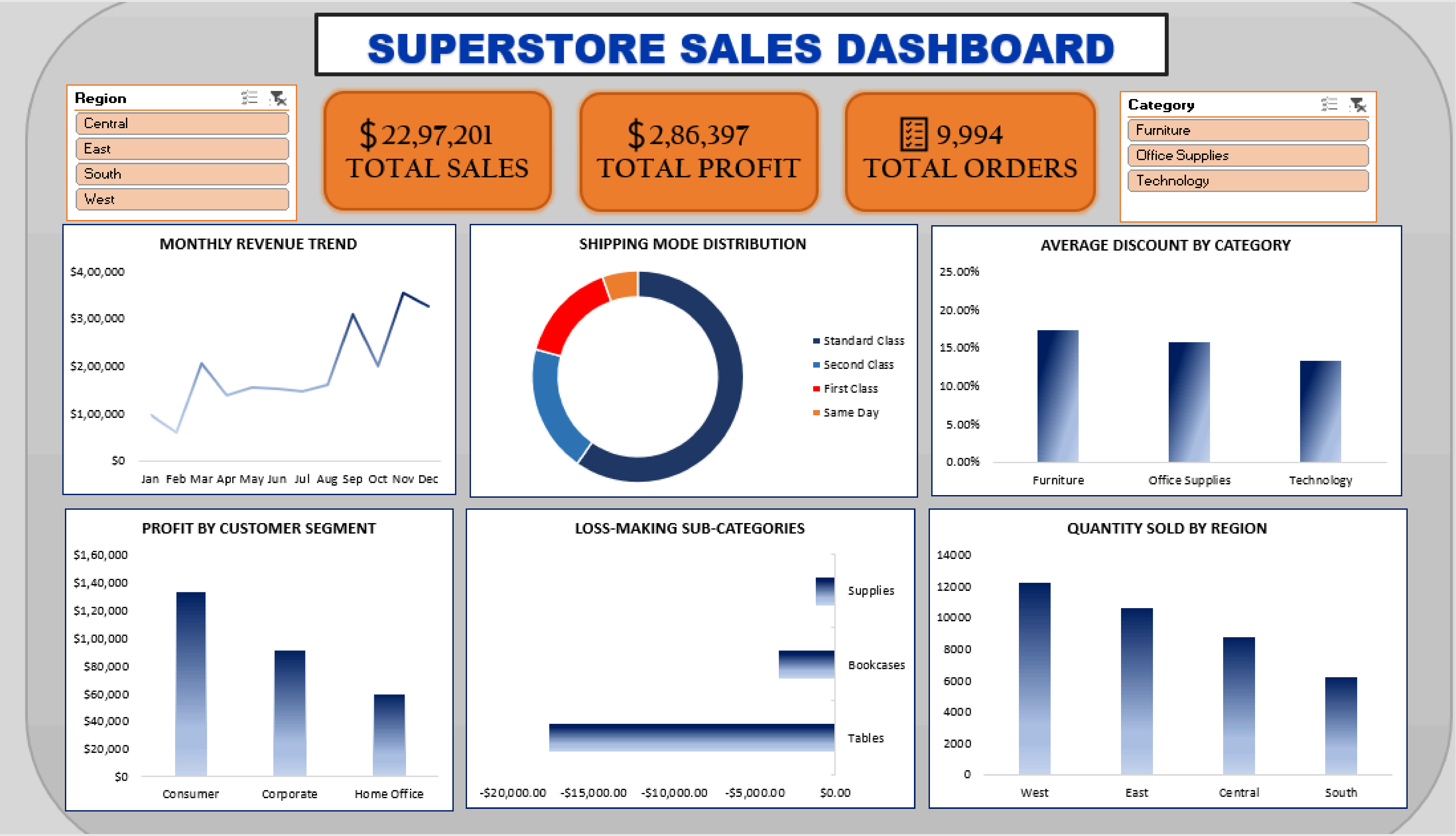
SHIPPING MODE DISTRIBUTION

Shipping Mode	Total Orders
Standard Class	5968
Second Class	1945
First Class	1538
Same Day	543
Grand Total	**9994**

QUANTITY SOLD BY REGION

Region	Total Quantity
West	12266
East	10618
Central	8780
South	6209
Grand Total	**37873**

EXCEL ANALYSIS



KEY INSIGHTS FROM THE PROJECT

- **Standard Class was the most frequently used shipping mode.**
- **Some sub-categories generated losses despite strong sales.**
- **Sales varied significantly across states and regions.**
- **Furniture category received higher average discounts.**
- **West region sold the highest quantity of products.**

SUGGESTIONS

- **Reduce excessive discounts on low-profit products.**
- **Improve strategies for loss-making sub-categories.**
- **Focus marketing efforts on high-performing regions.**
- **Optimize shipping methods for efficiency.**
- **Increase focus on profitable product categories.**

CONCLUSION

- The project successfully analyzed Superstore sales data using Python and Excel.

- The analysis helped identify:
  - sales trends
  - customer behavior
  - profitable regions
  - business opportunities

- Interactive dashboards and visualizations improved data interpretation and decision-making.

THANK YOU!